Adaptive AI-Based Phishing Detection System

Complete Project Understanding Guide

Read this before your supervisor meeting

Final Year Project · Computer Science · 2024/2025

#	Section	What You Will Learn
1	The Big Picture	What this project is, in the simplest terms
2	What Is Phishing?	The problem your system solves
3	The 5 Core Parts	Dataset, TF-IDF, Random Forest, Feature Extractor, Streamlit
4	Step-by-Step Flow	Exactly how a message goes from input to verdict
5	Performance Numbers	What accuracy, precision, recall, and the confusion matrix mean
6	The 5 Web Pages	Every page of the interface explained
7	Adaptive Learning	How your system gets smarter over time

8	Defense Q&A	10 questions your supervisor will ask — with full answers
9	Quick Summary	Everything in one reference table

1 What Is This Project?

The Big Picture

The Simplest Explanation Possible

Imagine you have a really smart friend who has read thousands of scam messages and thousands of normal, everyday messages. Now whenever you get a suspicious message, you show it to your friend and they immediately tell you — *'that is a scam, do not click it'* or *'that is perfectly safe.'*

That friend is your AI system. You built a computer program that reads any message or website link and tells you whether it is trying to steal your personal information. Everything else in this project is just the details of how it does that.

■ **This is not a basic checker.** It is an **ADAPTIVE** system — meaning it gets smarter every single time a user corrects it. The more it is used, the better and more accurate it becomes.

The Official One-Sentence Description

■ "An Adaptive AI-Based Phishing Detection System that uses Machine Learning (Random Forest classifier with TF-IDF text analysis) to automatically detect phishing messages and URLs in real time, with an adaptive learning feature that continuously improves the model from user feedback."

2 What Is Phishing?

The Problem Your System Solves

Phishing is when a criminal pretends to be someone you trust — your bank, PayPal, Netflix, Apple, or even the government — to trick you into giving them your password, credit card number, or personal details. They do this by sending fake messages and building fake websites that look identical to the real ones.

A Real Example of a Phishing Message:

> ■■ **URGENT:** Your PayPal account has been suspended. Verify immediately at: http://paypa1-secure.tk/verify
>
> *Notice 'paypa1' instead of 'paypal'. The criminal built a fake website that looks exactly like PayPal. If you click and type your password, they steal it instantly.*

Why Is It Called Phishing?

The name comes from fishing — criminals throw out bait (fake messages) and wait for victims to bite (click the link). They send millions of these messages and wait for anyone to fall for it. Your system is the net that catches them before they reach the victim.

Global Phishing Statistic	Figure
Annual financial damage from phishing (2022)	$52 Billion+
Increase in phishing attacks (2020–2023)	300%
Average lifespan of a phishing website	Less than 24 hours
Reason blacklists fail	Sites appear and disappear faster than lists update

3 The 5 Core Parts

How Every Piece Fits Together

Part 1 — The Dataset (dataset.csv): The AI's Textbook

Before a doctor can diagnose a disease, they need to have studied thousands of sick patients and thousands of healthy patients. Your AI needed exactly the same thing. You created a file with **280 messages** — some phishing, some legitimate — each labeled with a number:

Label Number	Meaning	Example Message
1	Phishing — DANGEROUS	URGENT: Verify your PayPal at http://paypa1.tk
0	Legitimate — SAFE	Hey, are we still meeting for lunch tomorrow?

■ This file is the AI's school textbook. It learns everything from these 280 examples. The more examples you add over time, the smarter it gets.

Part 2 — TF-IDF: The Word-to-Number Translator

Computers cannot read English. They only understand numbers. So you needed something that converts words into numbers the computer can process. That is what **TF-IDF (Term Frequency-Inverse Document Frequency)** does.

TF-IDF looks at every word in a message and gives it a score based on how important and unusual it is. Words that appear a lot in phishing messages but rarely in normal messages get HIGH scores. Common everyday words get LOW scores:

Word or Phrase	Score Given	Why
URGENT, verify, click here	HIGH danger score	Appears mostly in phishing messages
confirm, account suspended	HIGH danger score	Classic phishing language
meeting, invoice, report	LOW danger score	Normal everyday business words
Hey, thanks, attached	LOW danger score	Common in safe, friendly messages

Part 3 — Random Forest: The AI Brain (200 Judges Voting)

This is the actual AI model that makes the final decision. Here is the simplest possible way to understand it:

> ■ **Imagine you are not sure if a message is a scam.**
>
> So you ask **200 different people** to read it and each one votes: *'scam'* or *'safe'*. You count the votes at the end.
>
> If 185 out of 200 people say scam — you are 92.5% confident it is a scam. That exact percentage becomes your **Confidence Score**.
>
> That is Random Forest. It is **200 decision trees** (like 200 judges) all voting at the same time. The majority wins. This is why it is so accurate — one wrong judge does not ruin the result.

Reason It Was Chosen	Explanation
High Accuracy	Achieves 95-98% on phishing datasets in research literature
Handles Noise Well	200 trees voting means one bad tree cannot ruin the result
No Data Scaling Needed	Works on raw numbers — no complicated preprocessing
Shows Feature Importance	Reveals which words and patterns matter most
Fast Predictions	Returns a verdict in milliseconds
Beginner Friendly	Simple Python implementation, excellent documentation

Part 4 — The Feature Extractor: The Detective Checklist

Besides TF-IDF word analysis, your system also runs a detective checklist on every message — specifically looking at the URL and message patterns:

What It Checks	Example That Triggers It	Verdict
IP address in URL instead of name	http://192.168.1.1/login	Suspicious ■■
URL shortener hiding destination	bit.ly/abc123	Suspicious ■■
Urgent pressure language	URGENT, ACT NOW, VERIFY NOW	Suspicious ■■
Phishing keyword present	prize, winner, free, bitcoin, claim	Suspicious ■■

No HTTPS encryption	http:// instead of https://	Suspicious ■■
Hyphenated suspicious domain	paypal-secure-verify.tk	Suspicious ■■
Random numbers in URL	verify39271.tk/account	Suspicious ■■

Part 5 — Streamlit: The Face of the System

Parts 1 to 4 all happen invisibly in the background. What the user sees is the beautiful dark-themed chat interface. **Streamlit** is a Python library that takes your Python code and automatically turns it into a fully working website. You wrote Python. Streamlit handled all the web design — no web development skills were required.

4

How The System Works

Step-by-Step Flow From Input to Result

Step 1	**User Input:** The user pastes any suspicious message, email, or URL into the chat and clicks Send.
Step 2	**Detective Checklist:** The Feature Extractor runs through its checklist — checks for IP addresses, short URLs, urgent words, missing HTTPS, suspicious keywords. All findings recorded.
Step 3	**TF-IDF Conversion:** The message is converted from English words into a list of numbers (a vector). Words like 'urgent' and 'verify' get high scores. Normal words get low scores.
Step 4	**200 Trees Vote:** The numbered vector is sent to all 200 decision trees simultaneously. Each tree independently votes: phishing or safe.
Step 5	**Verdict + Confidence:** The majority vote becomes the verdict (PHISHING / SUSPICIOUS / SAFE). The percentage of trees that agreed becomes the Confidence Score.
Step 6	**Result Displayed:** The interface shows the verdict in color (red/amber/green), the confidence progress bar, and the specific reasons why the message was flagged.
Step 7	**User Feedback:** System asks: Was this correct? If the user says No, the correction is saved.
Step 8	**Adaptive Retraining:** When Retrain is clicked in Settings, all feedback is merged with training data and the model is rebuilt — smarter than before.

5 What The Numbers Mean

Performance Metrics in Plain English

Metric	Your Score	Plain English Meaning
Accuracy	100%	Out of 100 messages tested, it got all 100 correct. Like 100/100 on an exam.
Precision	100%	Every time it said 'phishing' — it was actually phishing. Zero false alarms.
Recall	100%	Every real phishing message in the test data was caught. Nothing slipped throu
F1-Score	100%	Combined score of Precision + Recall together. 100% means perfect on both.
CV Score	100% ± 0%	Tested 5 different ways on 5 data splits. Scored 100% every single time.

The Confusion Matrix — Explained Simply

A confusion matrix sounds scary but it is just a 2x2 grid showing the four possible outcomes of every prediction the AI makes:

	AI Predicted: SAFE	AI Predicted: PHISHING
Actually SAFE	True Negative = 26 ■ Correctly said safe	False Positive = 0 ■ Wrongly called phishing
Actually PHISHING	False Negative = 0 ■ Missed a real phishing	True Positive = 30 ■ Correctly caught phishing

■ Your system scored **0 False Positives** and **0 False Negatives** on the test set. This means it never wrongly accused a safe message AND never missed a real phishing attack.

6 The Web Interface

All 5 Pages Explained

■ Chat

The Main Page: This is where all the action happens. The user pastes any suspicious message, email, or URL and clicks Send. The AI returns: the verdict (PHISHING / SUSPICIOUS / SAFE) shown in red, amber, or green; a confidence score; a visual progress bar; and the exact reasons why the message was flagged. After each scan, it asks for feedback.

■ History

The Logbook: Every message ever scanned is saved here with the exact time, the verdict, and the confidence score. This works as a complete audit trail. In a real-world deployment, security teams would use this to review past incidents and identify patterns.

■ Statistics

The Dashboard: Shows four summary cards: Total Scans, Phishing Detected, Safe Messages, and Feedback Given. Below that are the model's performance scores (Accuracy, Precision, Recall, F1) and the full Confusion Matrix displayed with color-coded cells.

■■ Settings

Adaptive Learning Control: This is where retraining happens. After collecting user feedback corrections, click Retrain Model Now. The system merges all feedback with original training data and rebuilds the AI model automatically. Also has options to clear history and feedback data.

■■ About

Project Information Panel: A complete summary of the project — what it does, the technology stack used, performance scores, and version information. Perfect for showing during your academic defense or any live demonstration.

7 Adaptive Learning

How Your System Gets Smarter Over Time

This is the most impressive feature of your project and what separates it from a basic phishing checker. Most phishing tools are static — trained once and never updated. Yours continuously learns and improves.

> **The Security Guard Analogy:**
>
> Imagine a security guard on their first day at the gate. They do not recognize every criminal because criminals constantly change their appearance and tactics.
>
> But every time someone slips through and you tell the guard — 'that person was actually a criminal' — the guard remembers and updates their mental profile.
>
> Over weeks and months, the guard becomes incredibly sharp — not just from their original training, but from every single correction you gave them.
>
> **That is your adaptive learning system.**

The 5-Step Adaptive Learning Cycle:

Step	Action	What Happens
1	Scan	User pastes a message. AI gives a verdict and confidence score.
2	Question	System asks: 'Was this prediction correct?'
3	Correction	If user clicks 'No, incorrect' — the right label is saved to feedback.csv
4	Retrain	User goes to Settings and clicks 'Retrain Model Now'
5	Improve	Feedback merges with original data. Model rebuilt. AI is now smarter.

8 Defense Questions & Answers

Be Ready For Your Supervisor

■■ Read all of these carefully before your meeting. Your supervisor will likely ask at least 3–5 of these. The answers are written so you can say them naturally out loud.

Q: Explain your project in one sentence.

A: I built an AI-based system that uses machine learning — specifically a Random Forest classifier with TF-IDF text analysis — to automatically detect phishing messages and URLs in real time, with an adaptive learning feature that allows the model to continuously improve from user feedback.

Q: What is phishing and why is it a problem worth solving?

A: Phishing is a cyberattack where criminals impersonate trusted organizations to trick users into revealing sensitive information like passwords and bank details. It caused over $52 billion in damage globally in 2022 alone, and attacks increased by 300% between 2020 and 2023. Traditional blacklist-based defenses cannot keep up because phishing sites often last less than 24 hours before being replaced with new ones.

Q: Why did you choose Random Forest as your machine learning model?

A: Random Forest is an ensemble method that combines 200 decision trees using majority voting, making it robust against errors and resistant to overfitting. It handles imbalanced datasets well, provides feature importance scores for interpretability, requires no feature scaling, and achieves consistently high accuracy on text classification tasks. Research by Sahingoz et al. (2019) confirmed Random Forest outperformed six other algorithms on phishing URL detection with 97.98% accuracy.

Q: What is TF-IDF and why did you use it instead of something else?

A: TF-IDF stands for Term Frequency-Inverse Document Frequency. It converts text into numerical vectors by giving higher scores to words that appear frequently in one document but rarely across all documents. This means phishing-specific words like 'urgent', 'verify', and 'claim' receive high scores while common words like 'the' receive low scores. It is ideal for distinguishing phishing language from legitimate message language, and it is computationally efficient for real-time use.

Q: What is adaptive learning and how does it work in your system?

A: Adaptive learning is the ability of a system to improve over time based on new data and experience. In my system, when a user identifies an incorrect prediction, the correct label is saved to a feedback file. When the user triggers retraining from the Settings page, the system merges this feedback with the original training data and rebuilds the Random Forest model from scratch. This means the system evolves to handle new phishing tactics that emerge after the initial training — which is critical because phishing patterns change constantly.

Q: Can you explain what your confusion matrix shows?

A: My confusion matrix shows four outcomes. True Negatives: 26 safe messages correctly identified as safe. True Positives: 30 phishing messages correctly caught. False Positives: 0 — the system never wrongly accused a safe message as phishing. False Negatives: 0 — the system never missed a real phishing attack. This demonstrates perfect classification on the test set with zero false alarms and zero missed threats.

Q: What are the limitations of your system?

A: The system relies on text analysis and URL pattern matching. It does not render web pages or analyze JavaScript behavior, so sophisticated attacks using clean URLs with malicious page content could potentially evade it. The training dataset of 280 samples is effective for demonstration but would need significant expansion for production deployment. Real-time WHOIS domain lookups were excluded to keep the interface fast. The system also currently does not analyze email headers.

Q: What are the real-world applications of this project?

A: The system can be integrated into corporate email gateways to scan incoming messages before delivery. It can be embedded in antivirus software as a URL scanning module, deployed as a Chrome browser extension to warn users in real time, integrated into bank transaction security checks, or exposed as a REST API that other applications can query for phishing verdicts.

Q: How is your project different from existing solutions like Google Safe Browsing?

A: Existing solutions like Google Safe Browsing are primarily blacklist-based — they only block URLs that have already been reported. They fail completely on zero-day phishing attacks — newly created URLs not yet on any list. My system uses machine learning to detect patterns, so it can identify phishing attempts it has never seen before. It also includes adaptive learning for continuous improvement, which most commercial solutions do not offer to end users.

Q: What future improvements would you make to this system?

A: First, I would replace or augment Random Forest with a BERT-based transformer model for deeper semantic understanding of phishing language. Second, I would add visual similarity detection — taking screenshots and comparing them to known legitimate sites to catch pixel-perfect clones. Third, I would package the system as a Chrome browser extension for real-time in-browser protection. Fourth, I would add SHAP (SHapley Additive Explanations) values to show exactly which features drove each specific prediction — making the AI fully explainable to non-technical users.

9

Quick Reference Summary

Everything in One Place

Component	What It Is	Simple Analogy
dataset.csv	280 labeled training messages	The AI's school textbook
TF-IDF	Converts words to numbers	Translator from English to math
Random Forest	200 decision trees voting	200 judges deciding a case
Feature Extractor	URL and pattern checklist	Security guard's checklist at the gate
Streamlit	Turns Python code into a website	The suit and tie on the body
feedback.csv	Stores user corrections	The guard's notebook of lessons learned
Adaptive Retraining	Rebuilds model with new feedback	Guard getting smarter each week
Confidence Score	% of trees that agreed	Percentage of judges who voted the same
Confusion Matrix	4-cell prediction result grid	The exam marking sheet

"I built a system that uses Artificial Intelligence to protect people from phishing — one of the most damaging cybersecurity threats in the world. The system learns from its mistakes and gets smarter over time. That is not just a final year project. That is a real solution to a real problem."

Good luck with your supervisor meeting tomorrow. You built this. You understand it. Walk in with confidence. ■